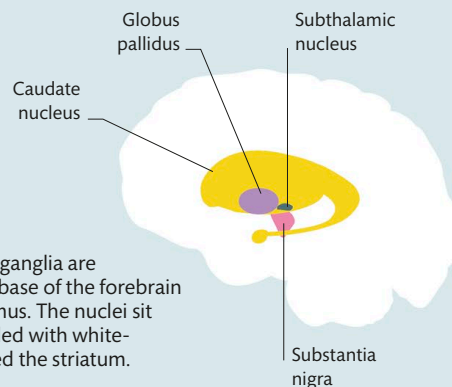


Nuclei of the Brain

In brain anatomy, a nucleus is a cluster of nerve cells that have a discernible set of functions and are connected to each other by tracts of white matter.

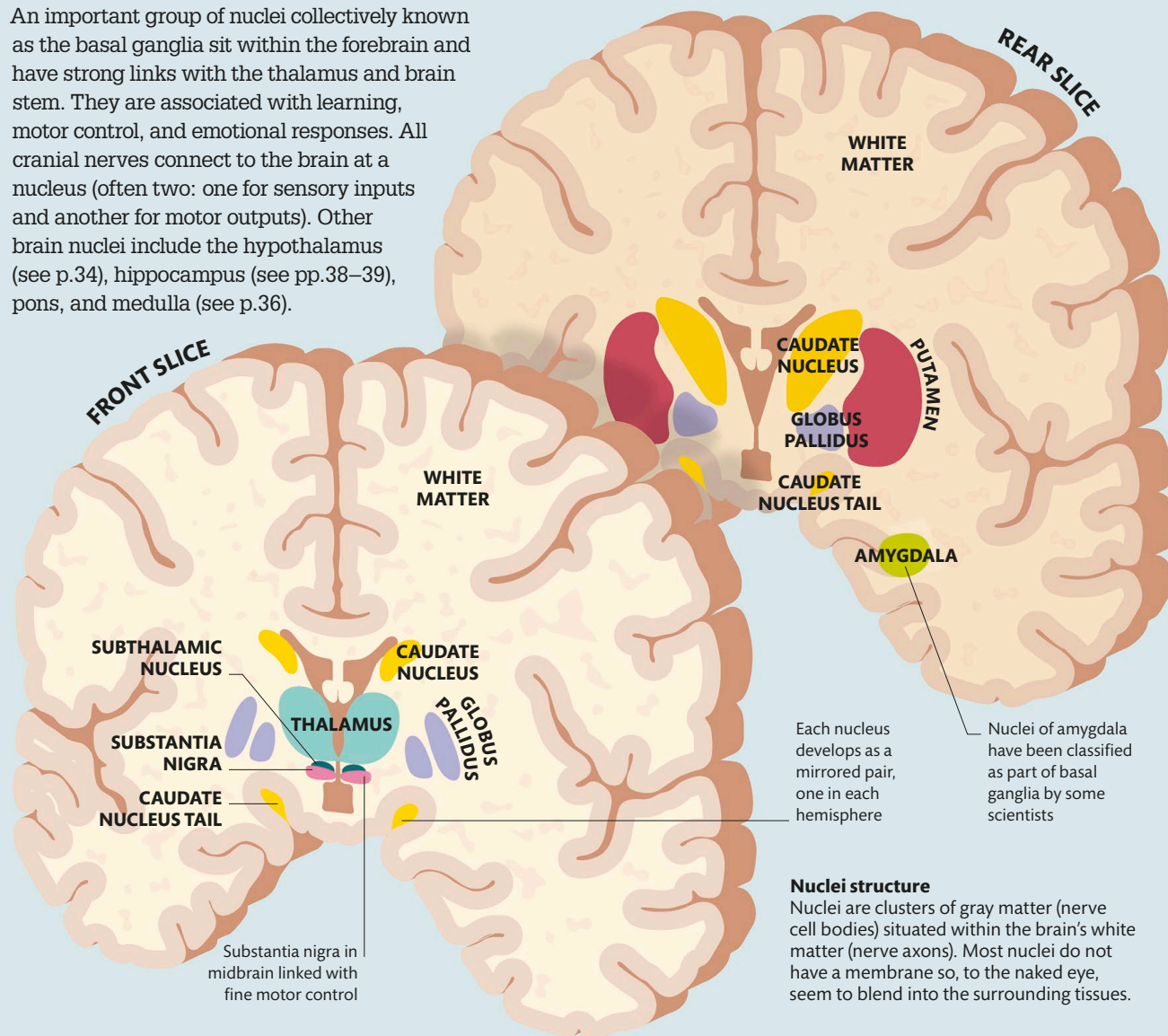
The basal ganglia and other nuclei

An important group of nuclei collectively known as the basal ganglia sit within the forebrain and have strong links with the thalamus and brain stem. They are associated with learning, motor control, and emotional responses. All cranial nerves connect to the brain at a nucleus (often two: one for sensory inputs and another for motor outputs). Other brain nuclei include the hypothalamus (see p.34), hippocampus (see pp.38–39), pons, and medulla (see p.36).



Central location

Most of the basal ganglia are positioned at the base of the forebrain around the thalamus. The nuclei sit within a region filled with white-matter tracts called the striatum.



Substantia nigra in midbrain linked with fine motor control

Each nucleus develops as a mirrored pair, one in each hemisphere

Nuclei of amygdala have been classified as part of basal ganglia by some scientists

Nuclei structure

Nuclei are clusters of gray matter (nerve cell bodies) situated within the brain's white matter (nerve axons). Most nuclei do not have a membrane so, to the naked eye, seem to blend into the surrounding tissues.